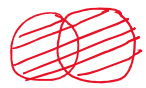
Lecture 3: Mutually Exhaustive & Exclusive Events

29 March 2026 10:40

Topics of This Lecture:

1. Mutually Exhaustive Events
2. Mutually Exclusive Events



$A \cup B = S$

1) Mutually Exhaustive Events:

If union of all events
cover all the elements in sample space
then those events are called mutually exhaustive events.

If $\boxed{A \cup B = S}$ then A & B are mutually exhaustive events.

Example:

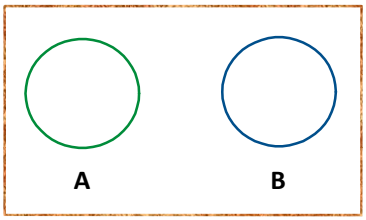
1. Tossing a coin $H \cup T = S$
2. Throwing a die $1 \cup 2 \cup 3 \cup 4 \cup 5 \cup 6 = S$
3. Drawing a card $1 \cup 2 \cup \ldots \cup 52 = S$

2) Mutually Exclusive Events:

If two events A & B cannot occur together
and intersection of two events A & B is zero
Then A & B are mutually exclusive events.

If $\boxed{A \cap B = 0}$ then A & B are mutually exhaustive events.

A = Shahrukh Khan
B = Sanjay Dutt



Event A & B are mutually Exclusive

$\boxed{A \cap B = 0}$

$\boxed{A \cap B = \phi}$

Example:

1. Tossing a coin
2. Throwing a die
3. Drawing a card

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- $A \cup B = |A| + |B| - |A \cap B|$ → dependent to each other
- $H \cup T = |H| + |T|$ → Independent of each other.

✓ $H \cup T = |H| + |T| = S$

✓ $H \cap T = 0$ nothing in common

→ mutually Exclusive
→ mutually Exhaustive } Both

- $A \cup B = S$ → Mutually Exhaustive events
- $A \cap B = 0$ → Mutually Exclusive events